

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 Scheme)

Course Code: CST 301**Course Name: FORMAL LANGUAGES AND AUTOMATA THEORY**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

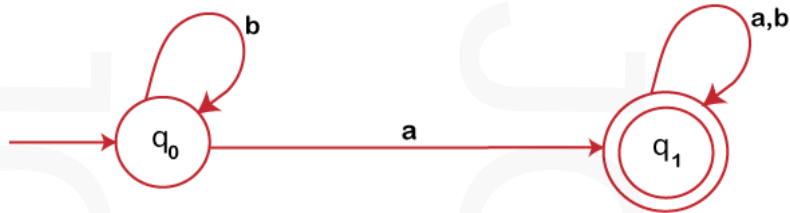
		Marks
1	Define ϵ -closure of a state? Explain with example.	3
2	Construct a DFA for strings in which first and last letters do not match. $\Sigma = \{a, b\}$	3
3	Generate regular expression for strings in which number of a's is a multiple of three. $\Sigma = \{a, b\}$	3
4	Explain any 3 closure properties of regular languages	3
5	With suitable example, explain about ambiguous grammar	3
6	State Myhill - Nerode Theorem	3
7	Whether DPDA and NPDA are equivalent? Justify your answer	3
8	Explain how CFGs can be converted to Chomsky Normal Form	3
9	Define Turing Machine	3
10	Differentiate between Recursive and Recursively Enumerable languages	3

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

- | | | |
|----|---|---|
| 11 | a) Prove that, if L is accepted by an ordinary NFA, there exist an equivalent ϵ -NFA that also accepts L | 7 |
| | b) Design an NFA (without ϵ -moves) for strings having substring ' bab '. Convert it into equivalent DFA. $\Sigma = \{a, b\}$ | 7 |
| 12 | a) Construct an ϵ -NFA for the language $L = \{0^n 1^m 2^p \mid n, m, p \geq 0\}$ and convert it into equivalent NFA without ϵ -transitions | 7 |
| | b) Design an NFA (without ϵ -moves) for strings with either consecutive zeros or consecutive ones. Obtain its corresponding DFA | 7 |

Module -2

- 13 a) Using pumping lemma, show that $L = \{ a^n b^n / n > 0 \}$ is not regular 7
 b) Develop equivalent automata for the Regular Expression $(a+b)^* aabb(a^*+bb)^*$ 7
- 14 a) Prove that for every Regular Expression 'R', there is an ϵ -NFA 'M' 7
 b) List out the rules for writing regular expressions. Convert the following DFA to its equivalent Regular Expression



Module -3

- 15 a) What is Greibach Normal Form (GNF)? Convert the following CFG to GNF 7
 $S \rightarrow AA / a, \quad A \rightarrow SS / b$
- b) a) Design CFG for the following languages 7
 (i) Palindromes over $\{a, b\}$
 (ii) Strings with more than 2 zeros. $\Sigma = \{0, 1\}$
 (iii) $(0+1)^*(01)^*(0+1)^*$
- 16 a) Minimize the following DFA using Myhill – Nerode theorem

	a	b
$\rightarrow q0$	q1	q2
q1	q1	q3
q2	q1	q2
q3	q1	*q4
*q4	q1	q2

- b) What is Chomsky Normal Form (CNF)? Convert the following productions to CNF. $S \rightarrow aSa / bSb / SS / \epsilon$ 7

Module -4

- 17 a) Prove that for every PDA accepted by final state, there exists an equivalent PDA accepted by empty stack. 7
- b) Design PDA for set of even length palindromes over $\{a, b\}$. Illustrate the working with suitable example 7
- 18 a) Design PDA for $L = \{x \in \{a, b\}^* / \#_a(x) = \#_b(x)\}$. 7
Here $\#_p(x)$ represents the number of occurrences of the symbol p in the string x
- b) Using pumping lemma for CFLs, show that $L = \{ww / w \in \{a, b\}^*\}$ is not context free. 7

Module -5

- 19 a) Design TM for $L = \{a^n b^m a^n / m, n > 0\}$. Illustrate the working with suitable example 7
- b) Explain Chomsky hierarchy for formal languages and evaluate various types 7
- 20 a) Design a TM to copy a string of a's and b's to the right side, leaving one blank symbol (b) in between. Assume that initially the input tape contains **bxb** and TM halts with **bxbx** as the tape content. $x \in \{a, b\}^*$ 7
- b) Prove that TM halting problem is undecidable 7
